

Research Assignment

DD / MM / YYYY

☐ MON

☐ TUE

☐ WED

☐ THU

☐ FRI

☐ SAT

☐ SUN

① What is Database

→ allows people and computer systems to store large amount of information

e.g.

→ hospitals can store patient information, appointments, medical records using database.

② What is Data Warehouse

→ is a central system that collects data from different sources and stores in a form that is prepared for analysis and reporting

e.g.

→ a supermarket collect sales information from its store into a data warehouse

* Database

→ run daily operations

* Data warehouse

* analyse historical and combined data

③ What is a Data Lake

→ large storage environment where an organization can keep large amounts of data in its original form.

* Data warehouse

→ stores cleaned and structured data that is ready for analysis

* Data Lake

→ Can store large amounts of raw data in many different formats

e.g.

Social media can store posts, photographs, videos and application logs in a data lake

④ What is a Data Mart

→ is a smaller, focused collection of data designed for a particular department, business function or group of users.

eg:

→ a company could have one central data warehouse containing information about the entire organisation.

⑤ Explain the difference between Structured, Semi-Structured and Unstructured Data

* Structured

→ data organised into a fixed format such as rows and columns

eg:

→ Excel spreadsheet / SQL table

* Semi-Structured

→ data that does not follow a traditional table but contains tags ~~or~~ keys

eg:

→ Customer records containing 'name', 'email' & 'age'

* Unstructured

→ data without a fixed structure

eg:

→ a photograph uploaded to a website

⑥ What is Data Modeling

→ the process of designing how data will be organised, connected and stored in a system

* Rational data model

→ Organised into tables consisting of rows and columns

eg:

→ Customer table can be connected to an Order table using Customer_ID

* Dimensional data model

→ it is used in data warehouse and analytics

e.g.

→ a shop could have a sales fact table connected to product, store and data dimensions.

① What is Data Mining

→ the process of examining large datasets to discover useful patterns, relationships, trends and insights

e.g.

* Banking

→ bank can analyse transaction patterns to identify unusual transactions that may indicate fraud

* Retail

→ supermarket can analyse purchases to discover that customers who buy coffee may also frequently buy biscuits

② What is a Database Management System

~~DBMS~~ - a software that allows users and applications to create, store, manage, update and retrieve data from database

* MySQL

→ popular relational database system

* PostgreSQL

→ powerful open-source relational database

* Oracle Database

→ enterprise database platform

* MongoDB

→ NoSQL database that store flexible document based data

⑨ What is ETL

* Extract

→ data is collected from different sources such as databases, spreadsheet or applications.

* Transform

→ data is cleaned and changed into a useful format

* Load

→ the transformed data is loaded into a destination such as a data warehouse

⑩ What are common data storage formats used in analytics

* CSV - Comma-Separated Values

→ Stores data in rows and columns using commas to separate values

→ use it for small datasets, spreadsheets and simple data exchange

* JSON - JavaScript Object Notation

→ stores data using key-value pairs and commonly used by APIs.

~~* Parquet~~

* Parquet

→ a column oriented storage format designed for efficient analytics

→ use it for large datasets and analytical systems such as data lakes

* Avro

→ a row-oriented data serialisation format that includes a schema describing the data

→ use it for data pipelines, distributed systems and streaming applications

Section 3: Artificial Intelligence & Machine Learning

MON TUE WED THU FRI SAT SUN

① What is Artificial Intelligence

→ technology that enables computers or machines to perform tasks that normally require human like intelligence.

* Narrow AI (weak AI)

→ is designed to perform specific tasks

e.g.

→ System that suggests movies based on your previous viewing behaviour

* General AI (strong AI)

→ a hypothetical form of AI that would have broad, human-like intelligence and be able to perform many different intellectual tasks

e.g.

→ a hypothetical machine capable of learning and solving a wide range of problems like human.

② What is Machine Learning?

→ a branch of AI where computers learn patterns from data and use those patterns to make predictions or decisions

* Supervised Learning

→ model learns from labelled data

* Unsupervised Learning

→ model works with data without predefined labels and tries to discover patterns or group

* Reinforcement Learning

→ an agent learn through interaction with an environment and receives rewards

(13) What is Deep Learning

→ a specialised part of machine learning that uses neural networks with multiple layers to learn complex patterns from data

* Neural network

→ a computing system inspired loosely by the biological brains process information.

→ it contains connected units called neurons or nodes organised into layers

(14) What is Natural Language Processing (NLP)

→ a field of AI that allows computers to work with human language in text or speech

* Voice assistants

→ understanding spoken questions

* Chatbots

→ understanding and responding to customers questions

* Translation tools

→ translating text from one language to another

(15) What is Large Language Model (LLM)

→ an AI model trained on very large amounts of texts so that it can understand patterns in language and generate text

* GPT

→ developed by OpenAI and used for conversational AI

* Gemini

→ developed by Google and used for conversational assistance, reasoning and working with different types of information

(16) What is Generative AI

→ a type of AI that can create new content based on patterns learned from training data

* ChatGPT

→ generates and works with text and can assist with analysis, coding and other tasks

* Microsoft Copilot

→ uses generative AI to assist users with tasks such as writing, summarizing and productivity

* Google Gemini

→ a generative AI assistant that can work with text and other forms of information

(17) What is Training Dataset

→ a collection of examples used to teach a machine-learning or AI model

→ if the data is incorrect or biased the model can produce inaccurate or unfair results

(18) What is Computer Vision

→ is an area of AI that allows computers to analysis and interpret images and videos

→ a computer does not see like humans, it process numerical representation of images and use learned patterns to identify objects, faces or shapes

eg:

- self-driving cars can use computer vision to identify roads, pedestrians and other cars
- hospitals can use computer vision to help analyse medical images such as X-rays or scans

Section: C: APIs, Tools & Developer Concepts

(19) What is an API

- a way for different software applications to communicate with each other and request information or services

eg

- at the restaurant, you tell the waiter what you want, the waiter sends the request to the kitchen and the chef (Kitchen) returns the results

(20) What is an API Key

- a secret credential used by an API provider to identify or ~~authenticate~~ authenticate an application or user
- if someone obtains your key may be able to use the associated services under your account
- should not be shared on social media, sent to people unnecessarily.

(21) What is a REST API

→ allows applications to communicate over HTTP using resources and standard HTTP methods

* GET

→ retrieve information

* POST

→ create / send new information

* PUT

→ update or replace information

* DELETE

→ remove information

(22) What is JSON

→ a lightweight format commonly used for storing and exchanging data, particularly between applications and APIs

```
{  
  "name": "Thabo",  
  "age": 25,  
  "city": "Portland"  
}
```

(23) What is Prompt Engineering

→ the practice of designing clear and effective instructions for an AI system

→ it is important because the quality and clarity of the instructions can affect the usefulness of the AI's response

eg.

* Poor prompt

→ make a dashboard

* Improved prompt

→ Create a dashboard plan for a coffee shop showing total sales, number of transactions, top-selling products, sales by store and sales by hour

(24) What is a MAP function

→ applies a function to each item in a collection such as a list or array and produce a new collection containing the results

eg.

* Python

→ numbers = [1, 2, 3, 4]

square = list(map(lambda x: x*x, numbers))
print(squares)(25) What is Cloud Computing

→ means using computing resources such as servers, storage, database and software over the internet instead of having to own and manage all the physical infrastructure

* IaaS→ ~~infrastructure as~~

→ provides infrastructure such as virtual machines and storage

eg.

→ Amazon EC2

* PaaS

→ provides a platform where developers can build and deploy applications without managing all the underlying infrastructure

eg:

→ Google App Engine

* SaaS

→ provides complete software that users access through the internet

eg:

→ Microsoft 365

②⑥ What is a Version Control System

→ a software that records changes to file over time

* Git

→ is one of the most widely used Version-Control Systems

→ allows developers and data analysts to track changes and return to previous versions

* Commit

→ is a saved record of changes

* Branch

→ a separate line of development that allows someone to work on changes without immediately changing the main version

②⑦ What is a Jupyter Notebook

→ an interactive environment where users can combine code, text, charts, Data and results.

→ popular with data and

→ popular with data analyst and data scientist because they can write code, analyse data and create visualisation

28) What is Python

→ a high-level programming language known for being relatively easy to read and widely used in data analytics, data science, AI and automation

* Pandas

→ used for data manipulation and analysis

* Numpy

→ used for numerical calculations and arrays

* Matplotlib

→ used for creating charts and visualisation

* Scikit Learn

→ used for machine learning tasks such as classification, regression and clustering

29) What is Data Privacy?

→ protecting people's personal information and ensuring that it is collected, stored, processed and shared responsibly

* General Data Protection Regulation (GDPR)

→ is a European Union regulation concerning the protection and processing of personal data

* Importantly, GDPR can also apply to organisations outside the EU if they offer goods & services to people in the EU or monitor their behaviour.

(30) What is Data Ethics?

- refers to the principles that guide the responsible collection, use, analysis and sharing data
- * Collecting personal data without proper Consent
 - collecting people's personal information can violate privacy and remove people's control over their information secretly
- * Using AI to discriminate
 - AI system used to hire and unfairly rejects applications from a particular group could create discrimination and unfair employment opportunities
- * Manipulating people using personal data
 - Company could use detailed information about people's behaviour to manipulate them into buying products or making decisions without their informed understanding

(31) What is Data Governance

- is the system of policies, rules, processes, responsibilities and standards used to manage an organisation's data properly.
- * Data ownership
 - who is responsible for particular data
- * Data quality
 - ensuring data is accurate and complete
- * Data access
 - deciding who can use particular information
- * Data privacy
 - protecting personal information
- * Data security
 - protecting information from unauthorised access.

32) What is Bias in AI

→ occurs when an AI system produces systematically unfair or inaccurate results for certain people or groups.

* incorrect labels

* Unrepresentative datasets

→ The AI could learn patterns from that historical bias and recommend similar candidates in the future.

* Loss of trust

* Discrimination

* Unfair hiring decisions

33) What are the key differences between a Data analyst, a Data Scientist, a Data Engineer and an AI Engineer

	skills	Tools
* Data Analyst	SQL, Excel, stats visualisation	Excel, SQL, Power BI, Tableau
* Data Scientist	Python, Statistics, ML	Python, Pandas, Scikit-learn, Jupyter
* Data Engineer	SQL, Python, database, Cloud	SQL, Python, Spark, Cloud platforms
* AI Engineer	Python, ML, API, AI framework	Python, TensorFlow / PyTorch, APIs, Cloud

(34) What is Business Intelligence (BI)

→ is the use of data, technology and analytical tools to help organisations understand their performance and make better decisions

* Microsoft Power BI

→ Can connect to different data sources and create interactive reports, dashboards and visualisations

* Tableau

→ is a visual analytics platform that helps users explore data and create interactive dashboards

* BI helps managers identify trends and make data-driven decisions.

(35) Which concept from this assignment did you find most interesting or surprising

→ Artificial Intelligence and Machine Learning the most interesting concepts

→ I thought AI was mainly about robots, but now I understand that AI is already used everyday.

(36) Which concept did you find most difficult to understand

→ Section B and C

→ I used ChatGPT to understand it better.

(37) How do you see Data and AI affecting your future career or everyday life

→ I believe that learning data analytics, SQL, Python, visualisation and AI will help me become more competitive in the future job market.

References

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